MTH 346/646 - Class 22 - 11/11

Fun fact: Let $E : y^2 = x^3 - x$ and let $p > 2$ be a prime number.

If $p \equiv 3 \pmod 4$, then $|E(\mathbb{F}_p)| = p + 1$.

If $p \equiv 1 \pmod 4$, then $p = a^2 + b^2$ for some integers a and b. Choose b to be even and $a \equiv 1 \pmod 4$ (negate a if necessary to make this happen). Then, $|E(\mathbb{F}_p)| = p + 1 - 2a$.

Last time: One more rank example, and then talking about cryptography.

Elliptic curve cryptography

The elliptic curve Diffie-Hellman algorithm works in a very similar way. Instead, a prime p is picked, and an elliptic curve $E/\mathbb{F}_p$ is used. The set $E(\mathbb{F}_p)$ is a group, and this group is used instead of $(\mathbb{Z}/p\mathbb{Z})^\times$.

A prime p, a curve E and a point $P \in E(\mathbb{F}_p)$ is agreed in advance. These are shared.

Alice chooses a random number α of size around p.

Bob chooses a random number β of size around p.

Alice computes αP and sends it to Bob. Bob computes βP and sends it to Alice.

Knowing βP and α, Alice computes $\alpha\beta P$ by adding βP to itself α times in the group law.

Knowing αP and β, Bob computes $\alpha\beta P$ by adding αP to itself β times in the group law.

The x-coordinate of $\alpha\beta P$ is the shared key.

The National Institute of Standards and Technology selected several particular elliptic curves to be used for cryptography. In 2013, concerns were raised that there might be a backdoor in NIST's P-256 curve, and Daniel Bernstein's Curve 25519 was used. The curve is

$$y^2 = x^3 + 486662x^2 + x$$

and the point P is chosen to have an x-coordinate of 9. The prime p used is $p = 2^{255} - 19$. This is used to transmit share a 128 bit AES key. (This curve was later "adopted" by NIST).

Elliptic curves over finite fields

Given a prime p, define

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } \gcd(a,p) = 1 \text{ and } \exists x \text{ so } x^2 \equiv a \pmod p \\ 0 & \text{if } p | a \\ -1 & \text{otherwise .} \end{cases}$$

In general, the number of square roots that a has mod p is $1 + \left(\frac{a}{p}\right)$.

Fact 1: We have $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \cdot \left(\frac{b}{p}\right)$.

Fact 2: There are $\frac{p-1}{2}$ numbers mod p with $\left(\frac{a}{p}\right) = 1$ and $\frac{p-1}{2}$ numbers mod p with $\left(\frac{a}{p}\right) = -1$.

If $E : y^2 = x^3 + ax + b$ is an elliptic curve mod p, then the number of points on $E/\mathbb{F}_p$ is

$$1 + \sum_{x=0}^{p-1} 1 + \left(\frac{x^3 + ax + b}{p}\right) = p + 1 + \sum_{x=0}^{p-1} \left(\frac{x^3 + ax + b}{p}\right).$$

You should think of $\left(\frac{x^3+ax+b}{p}\right)$ as a random number which is 1 half the time and -1 half the time.

Theorem (Hasse): If p is prime, we have

$$p + 1 - 2\sqrt{p} \leq |E(\mathbb{F}_p)| \leq p + 1 + 2\sqrt{p}.$$

The above theorem makes this plausible.

Elliptic Curve ElGamal Cryptosystem

This is used by the PGP (short for Pretty Good Privacy) system. Many national law enforcement agencies (the FBI, and the Italian police) have been unable to decrypt PGP-encrypted hard disks.

Alice wants to send a message to Bob.

(a) Bob chooses a prime q and an elliptic curve $E/\mathbb{F}_q$, where q is a large prime. He chooses a point $P \in E(\mathbb{F}_q)$ and a secret integer a. He compues aP.

(b) Bob sends q, E, P and aP to Alice.

(c) Alice expresses her message as a point $Q \in E(\mathbb{F}_q)$. She does this as follows. It must be that her message m is a few bits shorter than q. She takes her message m and appends K bits to the end of it. She tries to choose those last K bits so that $x = 2^K m + j$, where $0 \leq j \leq 2^K - 1$ has the property that $x^3 + ax + b$ is a square in $\mathbb{F}_p$. Since half of the elements in $\mathbb{F}_p$ are squares, the chances that she is successful (for some j) are $1/2^{K-1}$.

(d) Alice chooses a random integer b and computes $R_1 = bP$ and $R_2 = Q + b(aP)$. Alice sends R_1 and R_2 to Bob.

(e) Bob computes

$$R_2 - aR_1 = Q + b(aP) - a(bP) = Q.$$

This gives the point Q, which gets translated back to a message.

Why EC is better - Attacks on RSA and Diffie-Hellman

For the RSA algorithm, the security of the message relies on the inability of the world to ever factor $N = pq$. Starting around 2012, most places were using 2048-bit RSA keys (so N was about 620 digits), althogh a few were still using 1024-bit RSA keys.

The largest RSA key that has been factored (as far as we know) is RSA-768, a 232 digit number with 768 bits that was factored on 12/12/2009.

This uses the Number Field Sieve, the most effective general factorization algorithm for numbers that are products of a few primes. This is a very advanced factorization algorithm. (Implementing it completely is roughly equivalent to a Ph.D. in computer science.) It took a span of two years of computation to (using an average of 1000 computers) to complete this factorization. At the end, it was necessary to row reduce a sparse square matrix of size 192796550 (with about 144 nonzero entries per row). This took 119 calendar days.

There is an elliptic curve factorization algorithm where the running time depends not on the size of the number, but on the size of the prime factors that need to be found. This is the reason that having an RSA key that is a product of two large primes is the optimal choice. (More about this after break.)

A simpler version of the NFS - the quadratic sieve. (Invented in 1981)

(a) The goal is to factor n, and the easiest way to do this is to find x and y so that $x^2 \equiv y^2 \pmod{n}$.

(b) Choose a number B and make a list of the primes $p_1, \ldots, p_k$ less than B. (This is called the factor base. We say that a number is B-smooth if all of its prime factors are $\leq B$.)

(c) Find a function $y(x)$ so that $y(x) \bmod n$ is a square, and so that there are $\ell > k$ choices of x so that $y(x) \bmod n$ factors into the primes p_1 through p_k. Call these congruences relations. (Often $y(x) = (x + \lfloor\sqrt{n}\rfloor)^2 - n$ is used. The values of $y(x)$ are quite small if x is small, and so they are reasonably likely to be B-smooth.)

(d) Make an $k \times \ell$ matrix M where M_{ij} is 1 if the exponent of p_i in the jth relation is odd, and 0 otherwise. Finding a linear combination of the rows of M that is zero is equivalent to taking a product of the relations and getting a solution to $x^2 \equiv y^2 \pmod{n}$.

(e) Compute the null space of M (as a matrix over $\mathbb{F}_2$).

Using several different choices for $y(x)$ and parallelizing the search makes it more efficieint. This gives rise to the method "Multiple Polynomial Quadratic Sieve" (MPQS - 1987).

Story: In August 1977, right after RSA was made public, Martin Gardner wrote an article about it in Scientific American. Ron Rivest estimated that, it would take 40 quadrillion years to factor an 129-digit RSA modulus, and believed RSA was secure for this reason. He did not count on algorithmic improvements in integer factorization. In 1994, the MPQS was used to factor RSA-129 by having many computers (and a couple of fax machines too) sieve in parallel.

Example of the quadratic sieve: Let $n = 87463$. We take values of a that are close to $\sqrt{n} \approx 295$. We search for values of a for which a^2 is congruent to a number that is a square times a number all of whose prime factors are ≤ 29. We find that

$$\begin{aligned}
265^2 &\equiv 13^2(-1)\cdot 2\cdot 3\cdot 17\\
278^2 &\equiv 3^2(-1)\cdot 3\cdot 13\cdot 29\\
296^2 &\equiv 3^2\cdot 17\\
299^2 &\equiv 2\cdot 3\cdot 17\cdot 19\\
307^2 &\equiv 3^2\cdot 2\cdot 13\cdot 29
\end{aligned}$$

We want to find a "linear combination" of these congruences where the right hand side is a square. We put the exponents of -1, 2, 3, 13, 17, 19 and 29 into a matrix, one column for each congruence. We put a 1 in the matrix if the exponent on the number is odd, and a 0 if the exponent is even.

We get

$$M = \begin{bmatrix} 1 & 1 & 0 & 0 & 0\\ 1 & 0 & 0 & 1 & 1\\ 1 & 1 & 0 & 1 & 0\\ 0 & 1 & 0 & 0 & 1\\ 1 & 0 & 1 & 1 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}.$$

We find that the null space of M (with entries in $\mathbb{F}_2$, entries mod 2) is 1-dimension and spanned by $\begin{bmatrix}1\\1\\1\\0\\1\end{bmatrix}$. This tells us that if we multiply the first, second, third and fifth congruences, we get

$$(265\cdot 278\cdot 296\cdot 307)^2 \equiv 13^2\cdot 3^2\cdot 3^2\cdot 3^2\cdot (-1)^2\cdot 2^2\cdot 3^2\cdot 13^2\cdot 17^2\cdot 29^2$$

This gives $59411^2 \equiv 34757^2 \pmod{n}$. We find that $\gcd(n, 59411+34757) = 149$ and $\gcd(n, 59411-34757) = 587$ and so

$$87463 = 149\cdot 587.$$